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Sleeved stacked computer hardware chassis

Abstract

The present disclosure provides for a computing device. The device may include a chassis, a first component, and a second component. The chassis may include a sleeve, a removable panel detachably secured over an opening formed by the sleeve, a first tray extending from the removable panel into an interior of the sleeve, and a second tray extending from the removable panel into the interior of the sleeve. The first component may include a motherboard and be disposed on the first tray. The second component may include a power supply and be disposed on the second tray. The first tray may be oriented in parallel relative to the second tray. The first tray and the second tray may be vertically oriented.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) Not applicable.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable.

REFERENCE TO SEQUENCE LISTING OR COMPUTER PROGRAM LISTING APPENDIX

(3) Not Applicable.

BACKGROUND

(4) The present disclosure relates generally to computing devices. More particularly, the present disclosure relates to a chassis for computing devices or, in other words, mechanical structures implemented to arrange mechanical and/or electrical components of computing devices relative to one another.

(5) A personal computer is a stand-alone desktop computing device housed in a chassis. Such computing devices typically include a motherboard and central processing unit (“CPU”) positioned thereon, along with other circuitry and electrical components, such as a graphics processing unit (“GPU”) connected thereto. Such personal computers are typically provided with a chassis. Conventional chassis present numerous issues associated with maintenance, repair, and replacement of the various internal components of computing devices. For instance, in order to repair or replace a particular component of a computing device, a panel must be removed from the chassis, and in a manner that may be akin to maintenance on a particular component of a car engine, various intervening components must be dislodged or removed in order for a user to reach the component of concern.

(6) It would be advantageous to provide a computer chassis that allows for expedited maintenance, repair, and/or replacement of individual components of the computing device. What is needed, then, are improvements in apparatuses and methods for chassis of computing devices.

BRIEF SUMMARY

(7) This Brief Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(8) One aspect of the present disclosure is a computing device. The device may include a chassis, a first component, and a second component. The chassis may include a sleeve, a removable panel detachably secured over an opening formed by the sleeve, a first tray extending from the removable

panel into an interior of the sleeve, and a second tray extending from the removable panel into the interior of the sleeve. The first component may include a motherboard and be disposed on the first tray. The second component may include a power supply and be disposed on the second tray. The first tray may be oriented in parallel relative to the second tray. The first tray and the second tray may be vertically oriented. In some embodiments, the first tray is an acrylic mono-block.

(9) In some embodiments, the sleeve includes a vertical panel oriented parallel relative to the removable panel. The sleeve may further include a top panel, a bottom panel oriented parallel to the top panel, a left panel, and a right panel oriented parallel to the left panel, each extending between the vertical panel and the removable panel.

(10) In some embodiments, the first component further includes an add-on card positioned vertically above the motherboard. Additionally, the first and second component may be in electrical connection via a blind mate interface connector such as a Milli-Grid Blind Mate Interface (“BMI”) connector, or other suitable blind mate connectors. Furthermore, the second component may include a coolant supply. The coolant supply may be configured to transmit coolant between the first component and the second component via a quick-disconnect fitting.

(11) Another aspect of the present disclosure is chassis for a computing device. The chassis may include a sleeve, a removable panel detachably secured over an opening formed by the sleeve, a first tray extending from the removable panel into an interior of the sleeve, and a second tray extending from the removable panel into the interior of the sleeve. The first tray may be vertically oriented, the second tray may be vertically oriented, and the first tray may be oriented in parallel relative to the second tray.

(12) In some embodiments, the first tray is configured to secure a first component of a computing device, and the second tray is configured to secure a second component of a computing device. The first component may include a motherboard, and the second component may include a power supply.

(13) In some embodiments, the sleeve includes a vertical panel oriented parallel relative to the removable panel. The sleeve may further include a top panel, a bottom panel oriented parallel to the top panel, a left panel, and a right panel oriented parallel to the left panel, each extending between the vertical panel and the removable panel.

(14) In some embodiments, the first component further includes an add-on card positioned vertically above the motherboard. Additionally, the chassis may further include a BMI connector extending between the first tray and the second tray. The BMI connector may be configured to provide electrical communication between the first component and the second component. Furthermore, the second component may include a coolant supply. The chassis may further include a quick-disconnect fitting extending between the first tray and the second tray, such that the coolant supply may be configured to transmit coolant between the first component and the second component via the quick-disconnect fitting.

(15) Another aspect of the present disclosure is a method of providing a computing device. The method may include providing a sleeve, providing a first component, and providing a second component. The first component may include a motherboard, and the second component may include a power supply. The method may further include securing the first component to a first tray, and securing the second component to a second tray. The method may further include fixing the first tray and the second tray to a removable panel. The method may further include detachably securing the removable panel over an opening formed by the sleeve, such that the first tray and the second tray are inserted within an interior of the sleeve.

(16) In some embodiments, the sleeve includes a vertical panel oriented parallel relative to the removable panel. The sleeve may further include a top panel, a bottom panel oriented parallel to the top panel, a left panel, and a right panel oriented parallel to the left panel, each extending between the vertical panel and the removable panel.

(17) In some embodiments, the method further includes orienting the first tray in parallel relative to

the second tray. For example, the first tray and the second tray may each be vertically oriented.

(18) In some embodiments, the method further includes providing a BMI connector between the first component and the second component, such that the first component and the second component are in electrical communication via the BMI connector. In further embodiments, the second component includes a coolant supply. The method may further include providing a quick-disconnect fitting between the first component and the second component, such that the coolant supply is configured to transmit coolant between the first component and the second component via the quick-disconnect fitting.

(19) Numerous other objects, advantages and features of the present disclosure will be readily apparent to those of skill in the art upon a review of the following drawings and description of a preferred embodiment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front perspective view of a computing device, according to some embodiments of the present disclosure.
- (2) FIG. 2 is rear perspective view of the computing device of FIG. 1, according to some embodiments of the present disclosure.
- (3) FIG. 3 is a rear view of the computing device of FIG. 1, according to some embodiments of the present disclosure.
- (4) FIG. 4 is a rear perspective view of a computing device, according to further embodiments of the present disclosure.
- (5) FIG. 5 is front perspective view of the computing device of FIG. 4, according to some embodiments of the present disclosure.
- (6) FIG. 6 is a rear perspective view of a computing device, according to even further embodiments of the present disclosure.
- (7) FIG. 7 is front perspective view of the computing device of FIG. 6, according to some embodiments of the present disclosure.
- (8) FIG. 8 is a rear perspective view of an insert portion of a computing device, according to some embodiments of the present disclosure.
- (9) FIG. 9 is a rear perspective view of an insert portion of a computing device, according to further embodiments of the present disclosure.
- (10) FIG. 10 is a rear perspective view of an insert portion of a computing device, according to even further embodiments of the present disclosure.
- (11) FIG. 11 is a front perspective view of the insert portion of FIG. 10, according to some embodiments of the present disclosure.
- (12) FIG. 12 is a partial top perspective view of the insert portion of FIG. 10, according to some embodiments of the present disclosure.
- (13) FIG. 13 is a partial elevated view of the insert portion of FIG. 10, according to some embodiments of the present disclosure.
- (14) FIG. 14 is a front view of the insert portion of FIG. 10, according to some embodiments of the present disclosure.
- (15) FIG. 15 is a rear perspective view of an insert portion of a computing device, according to additional embodiments of the present disclosure.
- (16) FIG. 16 is a front perspective view of the insert portion of FIG. 15, according to some embodiments of the present disclosure.
- (17) FIG. 17 is a side view of the insert portion of FIG. 15, according to some embodiments of the present disclosure.

- (18) FIG. **18** is a front view of the insert portion of FIG. **15**, according to some embodiments of the present disclosure.
- (19) FIG. **19** is a rear perspective view of a chassis for a computing device, according to some embodiments of the present disclosure.
- (20) FIG. **20** is a rear perspective view of an insert frame for a computing device, according to some embodiments of the present disclosure.
- (21) FIG. **21** is a front perspective view of the insert frame of FIG. **20**, according to some embodiments of the present disclosure.
- (22) FIG. **22** is a front view of the insert frame of FIG. **20**, according to some embodiments of the present disclosure.
- (23) FIG. **23** is a side perspective view of the insert frame of FIG. **20**, according to some embodiments of the present disclosure.
- (24) FIG. **24** is a rear perspective view of the insert portion of FIG. **8** and a sleeve for a computing device, according to some embodiments of the present disclosure.
- (25) FIG. **25** is a rear perspective view of a sleeve for a computing device, according to further embodiments of the present disclosure.
- (26) FIG. **26** is a rear perspective view of a sleeve for a computing device, according to even further embodiments of the present disclosure.
- (27) FIG. **27** is a side view of an insert being inserted into a sleeve for a computing device, according to some embodiments of the present disclosure.
- (28) FIG. **28** is a front perspective view of the insert of FIG. **28** being inserted into the sleeve of FIG. **27**, according to some embodiments of the present disclosure.
- (29) FIG. **29** is a rear perspective view of the insert of FIG. **28** being inserted into the sleeve of FIG. **27**, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (30) While the making and using of various embodiments of the present invention are discussed in detail below, it should be appreciated that the present invention provides many applicable inventive concepts that are embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention and do not delimit the scope of the invention. Those of ordinary skill in the art will recognize numerous equivalents to the specific apparatus and methods described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.
- (31) In the drawings, not all reference numbers are included in each drawing, for the sake of clarity. In addition, positional terms such as “upper,” “lower,” “side,” “top,” “bottom,” etc. refer to the apparatus when in the orientation shown in the drawing, or as otherwise described. A person of skill in the art will recognize that the apparatus can assume different orientations when in use.
- (32) Referring now to FIGS. **1-7**, a computing device (“device”) **100** is shown, according to some embodiments of the present disclosure. As discussed in greater detail below, the device **100** may include a chassis **12** that is configured to house the various internal components of the device **100**. As provided for herein, such internal components may be internally organized in a first component **47** (e.g., a motherboard component, a component including a motherboard and associated electronics) and a second component **45** (e.g., a hardware component, a component including fans **51**, a power supply, cooling resources, etc.). The present disclosure provides for the device **100**, which may organize the first component **47** and the second component **45** in a stacked fashion that may be easily removed from and returned into a sleeve **19** formed by components of the chassis **12**. Advantageously, this sleeved, stacked, arrangement may facilitate improved ease-of-access to various components of the device **100**, thereby improving the repair, maintenance, and/or enhancement of the device **100**. Accordingly, the device may include the chassis **12**, the first component **47**, and the second component **45**.
- (33) For instance, FIGS. **1-3** depict the device **100** according to some embodiments of the present

disclosure; FIGS. 4 and 5 depict the device **100** according to alternative embodiments of the present disclosure; and FIGS. 6 and 7 depict the device **100** according to further embodiments of the present disclosure.

(34) The device **100** may include a motherboard **10** (e.g., a printed circuit board). For example, the motherboard **10** may include a central processing unit (“CPU”) and other associated electronics, such as control circuitry and other electronics. As a first example, device **100** may include a memory module disposed on the motherboard **10**. For instance, such memory module may be a memory card such as a random access memory (“RAM”) card. As a second example, the device **100** may include input/output (“IO”) connectors **21**, which may facilitate the connections of IO devices such as a video monitor, mouse, or printer to the control circuitry and other electronics of the device **100**.

(35) In some embodiments, the device **100** includes an add-on card **14** in communication with the motherboard **10**. Generally, the add-on card **14** (e.g., an expansion board, an expansion card, etc.) may be a second component that can be connected to the components of the motherboard **10** in order to enhance the functionality of the device **100**. For example, the add-on card **14** may be a graphics processing unit (“GPU”). The add-on card **14** may be positioned vertically above the motherboard **10**. For instance, an expansion connector **26** may be positioned on the motherboard **10** in order to facilitate such vertical connection between the motherboard **10** and the add-on card **14**. Generally, the device **100** may form the chassis **12** that includes a vertical panel **48** and a removable panel **18** oriented parallel to the vertical panel **48**. The chassis **12** may further include a top panel **50**, a bottom panel **52**, a right panel **54**, and a left panel **56** extending between the vertical panel **48** and the removable panel **18**. As discussed in greater detail below, the chassis **12** may be configured to house the electronic and working second components of the device **100**.

(36) In some embodiments, the removable panel **18** includes an interface **20** configured to secure the aforementioned one or more IO connectors **21** on the motherboard **10**. Additionally, the removable panel **18** may include an IO shield **23**. The IO shield **23** may be aligned with the add-on card **14**. The IO shield **23** may be configured to secure additional IO connectors disposed on the add-on card **14**. In some embodiments, particularly in cases where the add-on card **14** is a GPU, the IO shield **23** may include a vent. While the position of the IO shield **23** may be moved to any suitable location(s) on the chassis **12**, it should be appreciated that the parallel position of the add-on card **14** relative to the motherboard **10** may advantageously allow a free stream of air to pass through the chassis **12**, thus enhancing the cooling effects resulting from air travel throughout the chassis. Moreover, the chassis **12** may be reduced in terms of footprint size due to the vertically structured arrangement of the motherboard **10** and the add-on card **14** (e.g., the first component **47** discussed in greater detail below with reference to FIGS. 8-18).

(37) Referring now to FIGS. 8-18, the motherboard **10**, the removable panel **18**, and, in some embodiments, the add-on card **14**, may be components of a insert **25**. As discussed in greater detail below, the insert **25** may provide for the first component **47** and the second component **45** in a stacked fashion, which may be advantageous for the sleeved configuration of the chassis **12** as discussed herein. FIG. 8 depicts the insert **25** according to some embodiments of the present disclosure; FIG. 9 depicts the insert **25** according to other embodiments of the present disclosure; FIGS. 10-14 depict the insert **25** according to further embodiments of the present disclosure; and FIGS. 15-18 depict the insert **25** according to even further embodiments of the present disclosure. The insert **25** may be insertable and removable from the sleeve **19**, which is shown and discussed in further detail with reference to FIGS. 24-29.

(38) As shown with particular reference to FIGS. 8 and 9, and as suggested above, the add-on card **14** may be disposed on the expansion connector **26**, according to some embodiments of the present disclosure. The motherboard **10** and the add-on card **14** may be connected by the expansion connector **26** such that the add-on card **14** (or, more generally, the planar feature broadly defined by the shape of the add-on card **14**) is oriented in parallel relative to the motherboard **10** (or, more

generally, the planar feature broadly defined by the shape of the motherboard **10**). In further embodiments, the add-on card **14** is coplanar with the motherboard **10**. As shown, the add-on card **14** is positioned vertically above the motherboard **10**. Of course, in further embodiments of the present disclosure, the add-on card is positioned below the motherboard **10**, lateral to (e.g., to the left or right of) the motherboard **10**, or in some other suitable relative positioning.

(39) The insert **25** may include the removable panel **18**, a first tray **29**, and a second tray **31**. For instance, the removable panel **18** may be oriented vertically (e.g., in order to form the typical housing of the computing device **100**, parallel to a vertical axis **99** of the computing device **100**—shown with particular reference to FIG. **11**, etc.), and the first tray **29** and the second tray **31** may extend laterally therefrom (e.g., towards the vertical panel **48** of the fully assembled device **100**, along a longitudinal axis **97** of the computing device **100**—shown with particular reference to FIG. **11**, etc.). The motherboard **10**, along with various related electronics such as the add-on card **14**, may form the aforementioned first component **47** vertically oriented and disposed on the first tray **29** (thus, the first tray **29** may be considered a “motherboard tray”). In other words, the first component **47** may include the motherboard **10** and be disposed on the first tray **29**. The device **100** may further include the second component **45** separate from the first component **47**. Thus, the first component **47** and the second component **45** in a stacked fashion relative to a lateral axis **95** of the computing device **100** (shown with particular reference to FIG. **11**).

(40) The second component **45** may include hardware such as a power supply **41**, a coolant supply **49**, and radiators **22** (shown with reference to FIG. **11**, for example). Such components may be vertically oriented and disposed on the second tray **31** (thus, the second tray **31** may be considered a “hardware tray”). In other words, the second component **45** may include the power supply **41** (among other components) and be disposed on the second tray **31**. As shown with particular reference to FIGS. **13** and **17**, the power supply **41** on the second component **45** may be connected to the first component **47** via power connectors **43**, which may be easily accessible from the side of the insert **25**. Thus, when a user has removed the insert **25** from the sleeve **19** as discussed herein, the power connectors **43** may be easily accessed and disconnected in order to facilitate repair and/or maintenance operations.

(41) The first tray **29** may be oriented in parallel relative to the second tray **31**. Furthermore, the first tray **29** and the second tray **31** may be vertically oriented. Thus, the insert **25** may generally form two parallel components extending from the removable panel **18** along the longitudinal axis **97**, and separated by a distance along the lateral axis **95** of the computing device **100**: the first component **47** secured by the first tray **29**, and the second component **45** secured by the second tray **31**.

(42) As shown with particular reference to FIGS. **10-14**, the first tray **29** may be an acrylic “monoblock,” though which various fasteners may extend in order to secure the first component **47** (e.g., the motherboard **10**) to the first tray **29**. In other embodiments, and as shown with particular reference to FIGS. **8, 9**, and **15-18**, the first tray **29** may be a standard hardware panel. Depending on the implementation, the second tray **31** may be a standard hardware panel.

(43) As mentioned above, the first tray **29** and the second tray **31** may each extend from the removable panel **18**. In some embodiments, each of the first tray **29** and the second tray **31** are secured to the removable panel **18** via one or more mechanical fasteners. The first tray **29** may be secured in place relative to the second tray **31** along the lateral axis **95** of the device **100** by one or more mechanical set-offs **37** (shown with particular reference to FIG. **14**). In some embodiments, the mechanical set-offs **37** are secured to one of the first tray **29** or the second tray **31**, and make contact with the other un-connected tray in order to provide such mechanical spacing. In other embodiments, the mechanical set-offs are secured to both the first tray **29** and the second tray **31**.

(44) As shown with particular reference to FIGS. **12** and **14**, the first tray **29** may include a male or female component of a blind mate interface connector such as a Milli-Grid Blind Mate Interface (“BMI”) connector **59** (or other suitable blind mate connectors), and the second tray **31** may

include a corresponding component of the BMI connector 59. The BMI connector 59 may provide an easily removable connection between the electronic components of the first component 47 and the second component 45. In this sense, the first component 47 and the second component 45 may be in electrical connection via the BMI connector 59. Thus, the first tray 29 (and the first component 47 secured thereto) can be easily removed from the second tray 31 (and the second component 45 secured thereto) for maintenance and upgrade, and vice-versa. For instance, a user may simply remove the insert 25 from the sleeve 19 (as shown and discussed in greater detail with reference to FIGS. 27-29), release the mechanical fasteners securing the desired tray to the removable panel 18 (from outside the chassis 12), and operate the easily accessible BMI connector 59 in order to separate the desired component for isolated maintenance.

(45) Depending on the implementation, the insert 25 may include various components configured for maintaining the operating temperature of the device 100. In some embodiments, the second component 45 includes one or more radiators 22 (shown with particular reference to FIG. 11). Such radiators 22 may be configured to dispose of heat drawn by fans 51 (shown with particular reference to FIGS. 14, 16, and 17). Additionally, coolant may be circulated through the first component 47. As shown with particular reference to FIGS. 11 and 13, such coolant may be provided via a coolant supply 49. In some embodiments, coolant is transferred between the second component 45 and the first component 47 by one or more quick disconnect fittings 39 (shown with particular reference to FIGS. 11, 14, 16, 18). In this sense, the second component 45 may include the coolant supply 49, and the coolant supply 49 may be configured to transmit coolant between the first component 47 and the second component 45 via the quick-disconnect fittings 39. Thus, when the first component 47 or the second component 45 are removed for isolated maintenance as mentioned above, the quick disconnect fittings 39 may allow for such separation without the loss of any coolant presently in the circulation system between the first component 47 and the second component 45.

(46) Referring now to FIGS. 19-23, the insert 25 may include an insert frame 61. The insert frame 61 may make up the mechanical components required to secure the electronic and other working second components of the insert 25. Thus, the insert frame 61 may include the removable panel 18, the first tray 29, and the second tray 31. Thus, the insert frame 61 may form an interior component of the chassis 12. Accordingly, the chassis 12 may generally include the sleeve 19 and the insert frame 61 insertable therein. In other words, the chassis 12 may, in total, include the vertical panel 48, the removable panel 18, the top panel 50, the bottom panel 52, the right panel 54, the left panel 56, the first tray 29 extending from the removable panel 18, and the second tray 31 extending from the removable panel 18.

(47) Referring now to FIGS. 24-29, the sleeve 19 is shown, according to some embodiments of the present disclosure. As mentioned above, the insert 25 may be insertable and removable from the sleeve 19. The sleeve 19 may be formed as an open box by the vertical panel 48, the top panel 50, the bottom panel 52, the right panel 54, and the left panel 56. For instance, rear edges of the top panel 50, the bottom panel 52, the right panel 54, and the left panel 56 may form an opening 60 of the sleeve 19.

(48) The removable panel 18 may be removably secured to (e.g., over the opening 60 of) the sleeve 19. In this sense, the sleeve 19 may include the vertical panel 48 oriented parallel to the removable panel 18; as well as the top panel 50, the bottom panel 52 oriented parallel to the top panel 50, the left panel 56, and the right panel 54 oriented parallel to the left panel 56, each extending between the vertical panel 48 and the removable panel 18. Thus, the chassis 12 may include the removable panel 18 detachably secured over the opening 60 formed by the sleeve 19. Furthermore, the chassis 12 may include the first tray 29 extending from the removable panel 18 into an interior 69 (shown with particular reference to FIG. 29) of the sleeve 19, and the second tray 31 extending from the removable panel 18 into the interior 69 of the sleeve 19.

(49) When the removable panel 18 is secured to the sleeve 19 as shown, the chassis 12 may form a

box as suggested above. Owing to the aforementioned box shape, the rear and vertical panels **18**, **48** may be parallel (or substantially parallel), the top and bottom panels **50**, **52** may be parallel (or substantially parallel), and the left and right panels **54**, **56** may be parallel (or substantially parallel). Thus, the chassis **12** may form a box around the insert **25**. Accordingly, and with additional reference to FIGS. **1-7**, an outer portion of the chassis **12** may be provided when the insert **25** is inserted through the opening **60**, such that the insert **25** is disposed in the sleeve **19**, thus joining the removable panel **18** with the sleeve **19** and housing the electronic components of the insert **25** within the housing formed by the removable panel **18** and the sleeve **19**.

(50) Referring now to FIGS. **27-29**, the device **100** is shown to depict the removal of the removable insert **25** from, or insertion of the same into, the sleeve **19**, according to some embodiments of the present disclosure. As suggested above, the sleeve **19** may be arranged such that the opening **60** is left where the removable panel **18** would otherwise be. The dimensions of the removable insert **25** may be arranged such that the removable insert **25** is insertable within the sleeve **19**. Generally, this advantageously provides access to the motherboard **10** in a manner that may not require significant steps or manual labor, such as disassembling a side panel, as may be the case in conventional computing devices. As shown with particular reference to FIG. **27**, the insert **25** may be moved in a direction **58** (e.g., along the lateral axis **97** of the device **100**) to be inserted into or removed from the sleeve **19**. Accordingly, due at least in part to the stacked configuration of the first component **47** and the second component **45** of the insert **25**, the insert **25** may be inserted into the sleeve **19**, thereby forming a sleeved configuration for the chassis **12**.

(51) As discussed herein, the present disclosure further provides for a method of providing the device **100**. For instance, such method may include providing the sleeve **19**, providing the first component **47**, and providing the second component **45**. The first component **47** may include the motherboard **10**, and the second component **45** may include the power supply **41**. The method may further include securing the first component **47** to the first tray **29** and securing the second component **45** to the second tray **31**. The method may further include fixing the first tray **29** and the second tray **31** to the removable panel **18**. The method may further include detachably securing the removable panel **18** over the opening **60** formed by the sleeve **19**, such that the first tray and **29** the second tray **31** are inserted within the interior **69** of the sleeve **19**.

(52) Thus, although there have been described particular embodiments of the present invention of a new and useful SLEEVED STACKED COMPUTER HARDWARE CHASSIS, it is not intended that such references to particular embodiments be construed as limitations upon the scope of this invention.

Claims

1. A computing device, comprising: a chassis including a sleeve, a removable panel detachably secured over an opening formed by the sleeve, a first tray secured to and laterally extending from the removable panel into an interior of the sleeve, and a second tray secured to and laterally extending from the removable panel into the interior of the sleeve; a first component disposed on the first tray, the first component including a motherboard; a second component disposed on the second tray, the second component including one or more fans, wherein the removable panel includes an interface securing one or more input/output connectors on the motherboard; and an insert including the removable panel, the first tray, and the second tray, wherein the sleeve includes a vertical panel oriented parallel to the removable panel when the insert is inserted into the sleeve and covers the opening, and wherein the sleeve further includes a top panel, a bottom panel oriented parallel to the top panel, a left panel, and a right panel oriented parallel to the left panel, each extending between the vertical panel and the removable panel.
2. The device of claim 1, wherein the first tray is oriented in parallel relative to the second tray.
3. The device of claim 2, wherein the first tray and the second tray are each vertically oriented.

4. The device of claim 3, wherein the first component further includes an add-on card positioned vertically above the motherboard.
5. The device of claim 4, wherein the first component and the second component are in electrical communication via a blind mate interface (BMI) connector.
6. The device of claim 4, wherein the second component includes a coolant supply, and wherein the coolant supply is configured to transmit coolant between the first component and the second component via a quick-disconnect fitting.
7. The device of claim 4, wherein the first tray is an acrylic mono-block.
8. A chassis for a computing device, the chassis comprising: a sleeve; a removable panel detachably secured over an opening formed by the sleeve; a first tray secured to and laterally extending from the removable panel into an interior of the sleeve; a second tray secured to and laterally extending from the removable panel into the interior of the sleeve, wherein the first tray is oriented in parallel relative to the second tray, wherein the first tray is configured to secure a first component of a computing device, the first component including a motherboard, wherein the second tray is configured to secure a second component of the computing device, the second component including one or more fans, and wherein the removable panel includes an interface securing one or more input/output connectors on the motherboard; and an insert including the removable panel, the first tray, and the second tray, wherein the sleeve includes a vertical panel oriented parallel to the removable panel when the insert is inserted into the sleeve and covers the opening, and wherein the sleeve further includes a top panel, a bottom panel oriented parallel to the top panel, a left panel, and a right panel oriented parallel to the left panel, each extending between the vertical panel and the removable panel.
9. The chassis of claim 8, wherein the first tray and the second tray are each vertically oriented.
10. The chassis of claim 9, wherein the first component further includes an add-on card configured to be positioned vertically above the motherboard.
11. The chassis of claim 9, further comprising a blind mate interface (BMI) connector extending between the first tray and the second tray, wherein the BMI connector is configured to provide electrical communication between the first component and the second component.
12. The chassis of claim 9, wherein the first tray is an acrylic mono-block.
13. A method of providing a computing device, the method comprising: providing a sleeve; providing a first component, the first component including a motherboard; providing a second component, the second component including one or more fans; securing the first component to a first tray; securing the second component to a second tray; fixing the first tray and the second tray to a removable panel; and inserting an insert into the sleeve and detachably securing the removable panel over an opening formed by the sleeve, such that each of the first tray and the second tray laterally extend from the removable panel and are inserted within an interior of the sleeve, wherein the removable panel includes an interface securing one or more input/output connectors on the motherboard, wherein the insert includes the removable panel, the first tray, and the second tray, wherein the sleeve includes a vertical panel oriented parallel to the removable panel when the insert is inserted into the sleeve and covers the opening, and wherein the sleeve further includes a top panel, a bottom panel oriented parallel to the top panel, a left panel, and a right panel oriented parallel to the left panel, each extending between the vertical panel and the removable panel.
14. The method of claim 13, further comprising orienting the first tray in parallel relative to the second tray.
15. The method of claim 13, wherein the first tray and the second tray are each vertically oriented.
16. The method of claim 15, further comprising providing a blind mate interface (BMI) connector between the first component and the second component, such that the first component and the second component are in electrical communication via the BMI connector.
17. The device of claim 4, wherein the second component further includes a coolant supply, and wherein the method further comprises providing a quick-disconnect fitting between the first

component and the second component, such that the coolant supply is configured to transmit coolant between the first component and the second component via the quick-disconnect fitting.
